

数字图像处理

Digital Image Processing

浙江大学信电学院

2021春夏

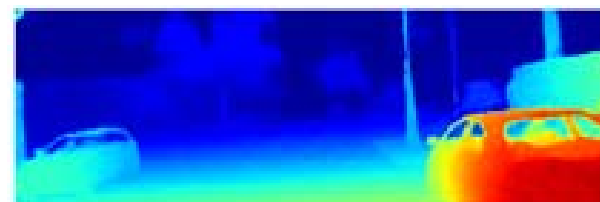
课程概要

- 代码: [67190030](#)
- 名称: 数字图像处理 *Digital Image Processing*
- 学分: 3.0
- 学时: 48
- 目标:
 - 学习数字图像处理的基本概念和理论知识
 - 视觉信息的处理与利用
 - 培养利用图像处理技术解决实际问题的能力

教学联络信息

- 主讲教师
 - 项志宇教授, 13989452718/662718
 - xiangzy@zju.edu.cn
 - 个人主页: <https://person.zju.edu.cn/xiangzy>
- 助教
 - 乔程昱, 博士生
 - 3140104427@zju.edu.cn
 - 每周三下午2:00-5:00答疑, 信电楼317

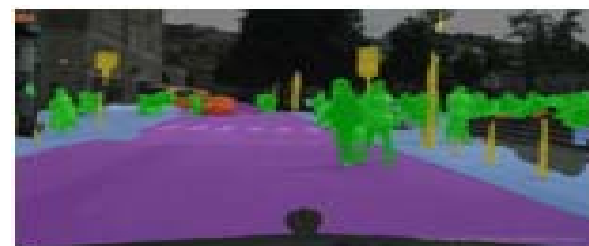
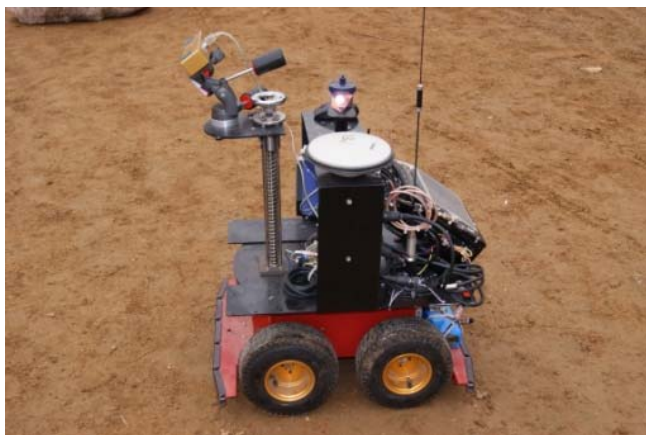
我们的**机器视觉与导航实验室**@ZJU



深度补全



目标检测



语义分割

课程教材与参考书

■ 教材

- 《数字图像处理（第四版）》 **Gonzalez**, Richard E. Woods 著，阮秋琦、阮宇智等译，电子工业出版社，2020年
- 教材参考网站： www.imageprocessingplace.com

■ 参考书

- Rafael C. **Gonzalez**, Richard E. Woods, "Digital Image Processing", Third Edition, Prentice Hall, 2010
- Kenneth R. **Castleman**, "Digital Image Processing", Prentice Hall, 1996
- 《数字图像处理（Digital Image Processing）》 K.R. **Castleman** 著，朱志刚等译
- William K. **Pratt**, "Digital Image Processing: PIKS Inside", Third Edition. John Wiley & Sons, Inc. 2001
- 《图像工程》， **章毓晋**编著，清华大学出版社

2018年6月冈萨雷斯教授访问并授课

学术报告
ACADEMIC LECTURE

题目: Digital Image Processing
Research & Commercialization

时间: 6月14日上午9:00
地点: 信电楼215会议室
报告人: Prof. Rafael C. Gonzalez (IEEE Fellow)

Abstract: Digital image processing is a mainstream field of research and application in areas ranging from medicine, biotechnology, and environmental monitoring, to industrial inspection, biometrics, robotics, and visual aspects of machine learning and artificial intelligence. This talk is an overview of the field, its history, and major current applications. Principles of how image processing technology is commercialized will also be discussed and illustrated with products that have resulted from academic research, and which have been commercially successful. Because of the recent focus of the Chinese government on innovation and its strategic importance in building a modernized economy, this talk should also be of interest to individuals working in research areas that have the potential for becoming innovative technologies with commercial applications.

Biography: R. C. Gonzalez received the B.S.E.E. degree from the University of Miami in 1965 and the M.E. and Ph.D. degrees in electrical engineering from the University of Florida, Gainesville, in 1967 and 1970, respectively. His academic honors for work in these fields include the 1977 UTK College of Engineering Faculty Achievement Award; the 1978 UTK Chancellor's Research Scholar Award; the 1980 Magnavox Engineering Professor Award; and the 1980 M.E. Brooks Distinguished Professor Award. In 1981 he became an IBM Professor at the University of Tennessee and in 1984 he was named Alumni Distinguished Service Professor there. He was awarded a Distinguished Alumnus Award by the University of Miami in 1985, the Phi Kappa Phi Scholar Award in 1986, and the University of Tennessee's Nathan W. Dougherty Award for Excellence in Engineering in 1992.

浙江大学 信息与电子工程学院

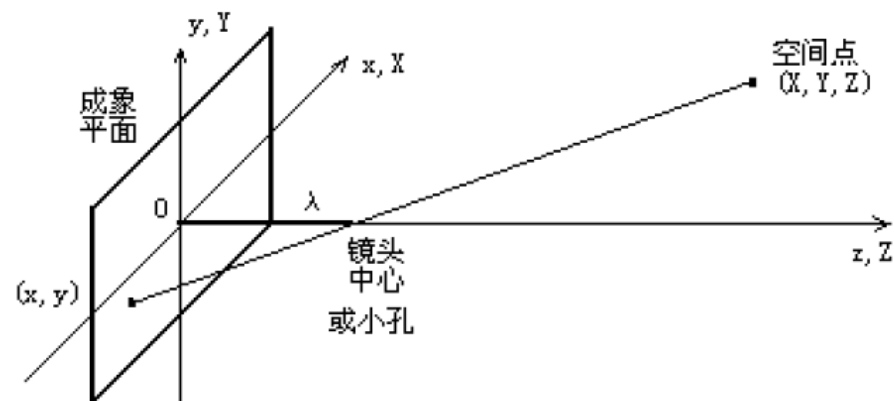


2021/2/26

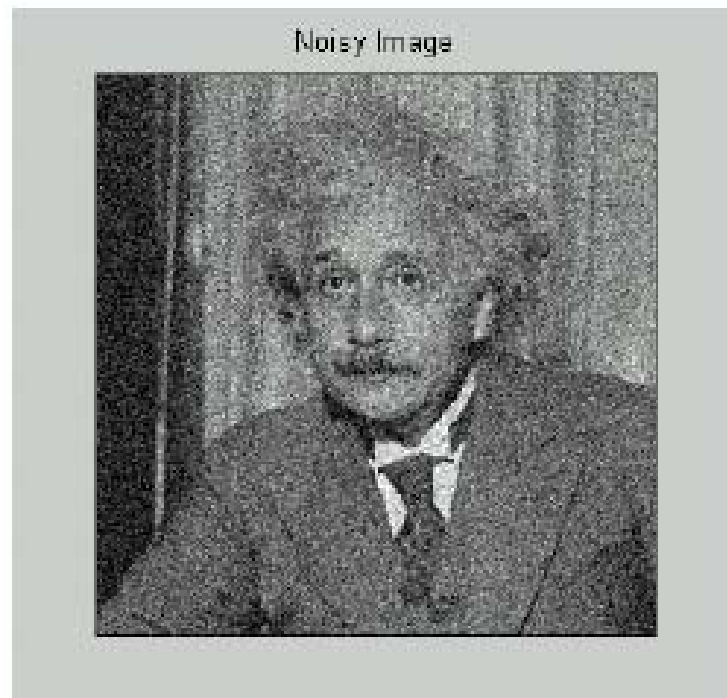
课程内容

- 1 绪论 (2.0学时)
- 2 数字图像基础 (4.0学时)
- 3 灰度变换与空间滤波 (6.0学时)
- 4 频率域滤波 (6.0学时)
- 5 图像复原与重建 (5.0学时)
- 6 彩色图像处理 (3.0学时)
- 7 形态学图像处理 (5.0学时)
- 8 图像分割 (6.0学时)
- 9 表示和描述 (5.0学时)
- 10 目标识别 (2.0学时)
- 11 图像处理c++程序设计框架入门 (1学时)
- 12 图像处理综合设计成果展示 (3.0学时)

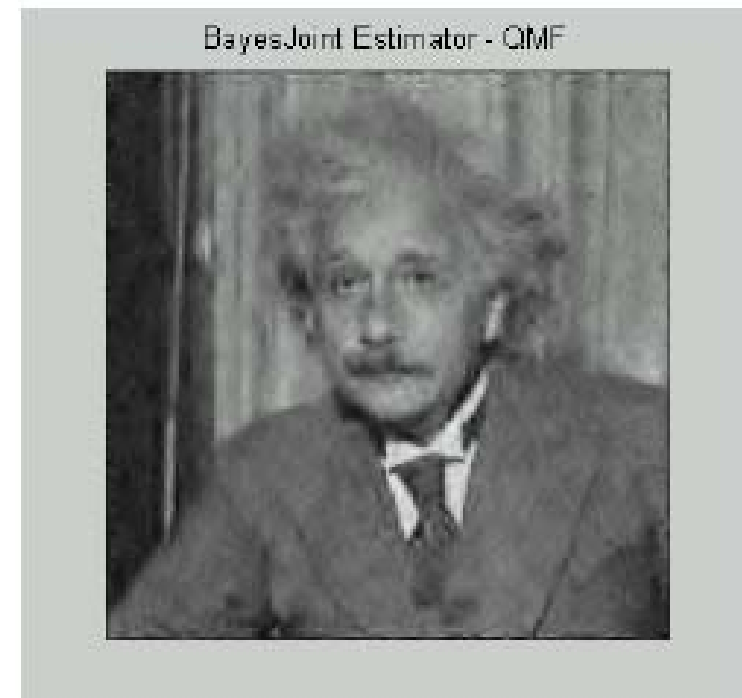
数字图像基础-视觉与图像生成



灰度变换与空间滤波



Degraded image

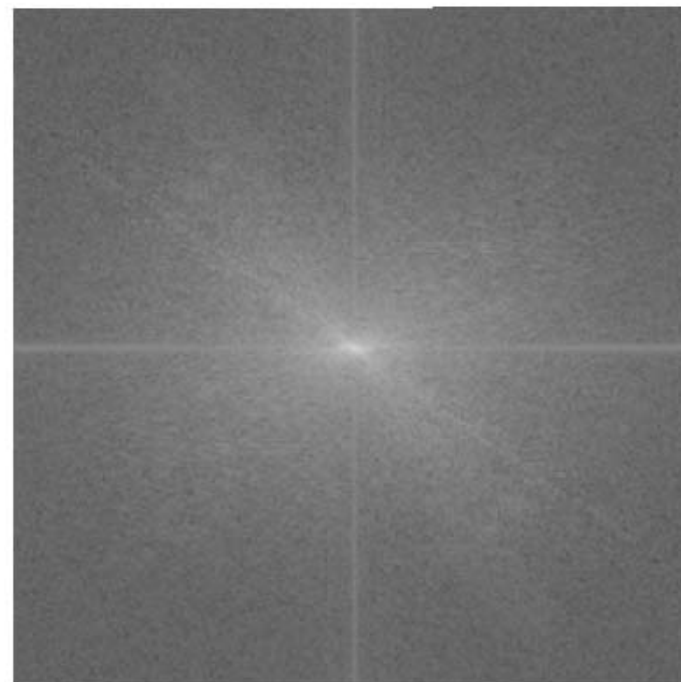


Noise-reduced image

二维空域-频域变换与滤波



(a)



(b)

图 6-4 Lenna 图及其频谱图

图像增强

Color photo enhancement



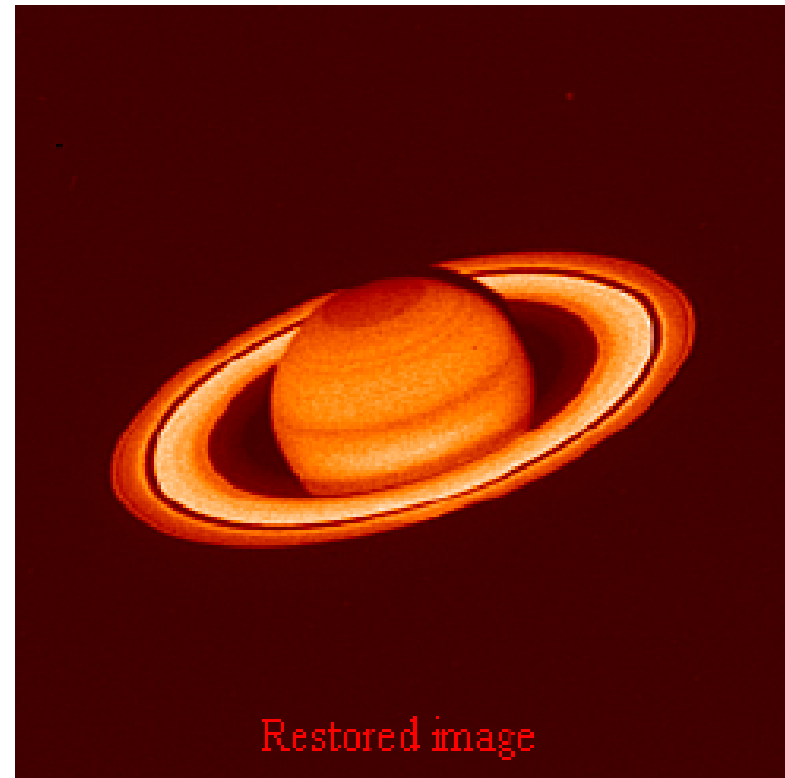
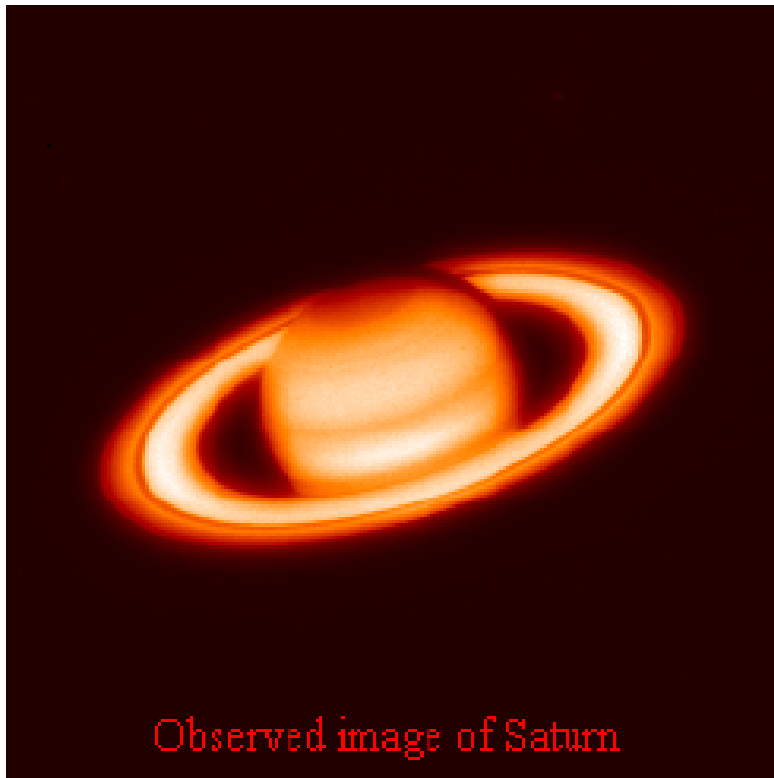
Original



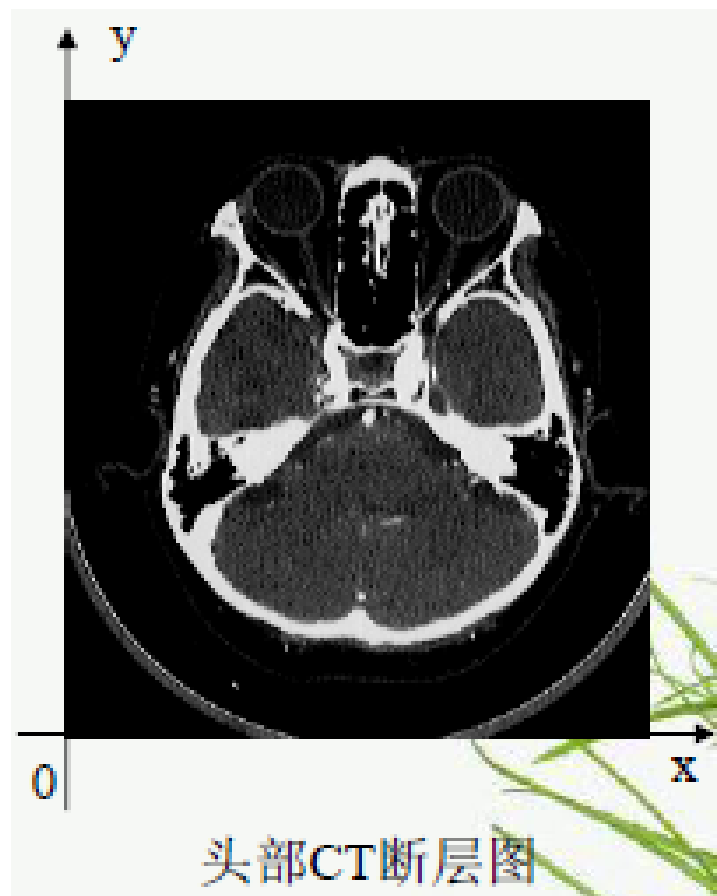
Automatic Enhancement

图像复原

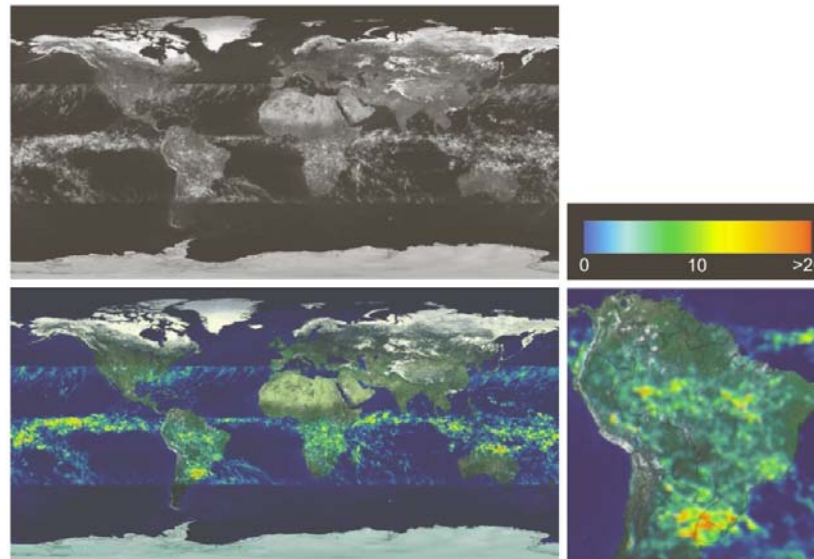
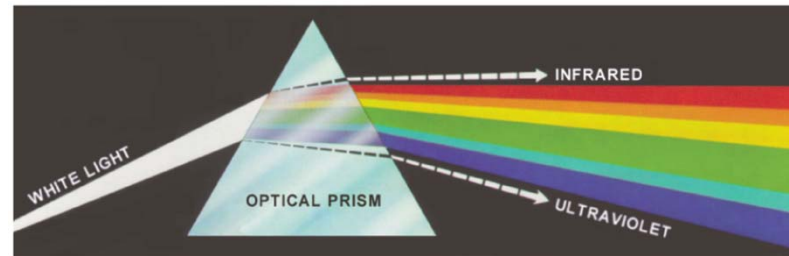
Restoration of image from Hubble Space Telescope



图像投影重建



彩色图像处理



数学形态学



(a) A



(b) $A \ominus B$



(c) $(A \ominus B) \oplus B = A \circ B$



(d) $(A \odot B) \ominus B$



(e) $[(A \odot B) \ominus B] \ominus B = (A \odot B) \cdot B$

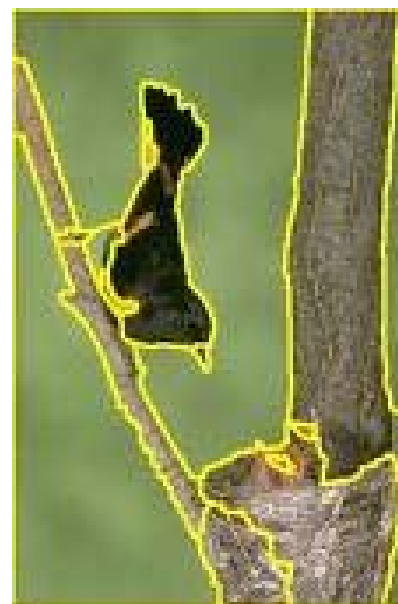
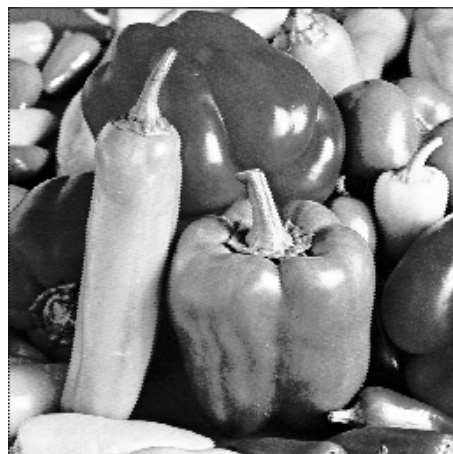
二值噪声滤除

1	1	1
1	1	1
1	1	1

B

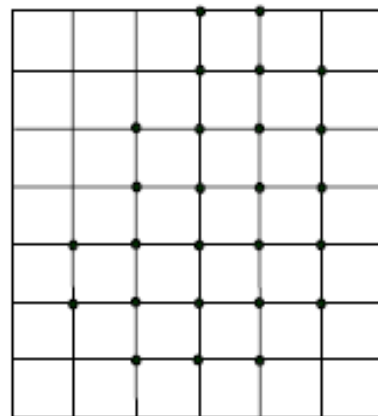
(f) 结构元

图像分割

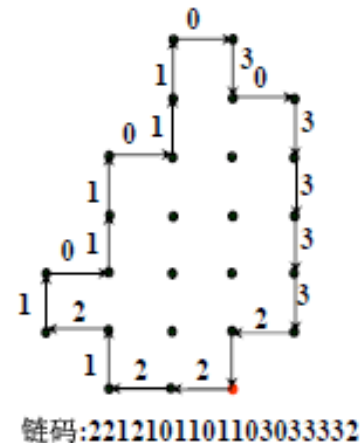


表示与描述

- 分割出来的目标，如何表达、描述？
 - 边界
 - 区域
 - 关系



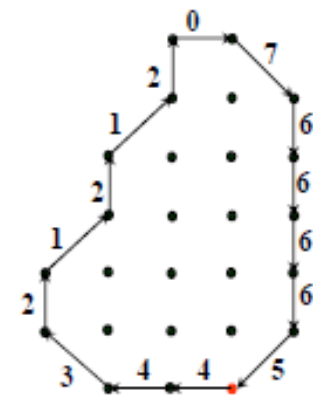
(a)分割后的目标像素



链码:221210110110303332

(b) 4-方向链码

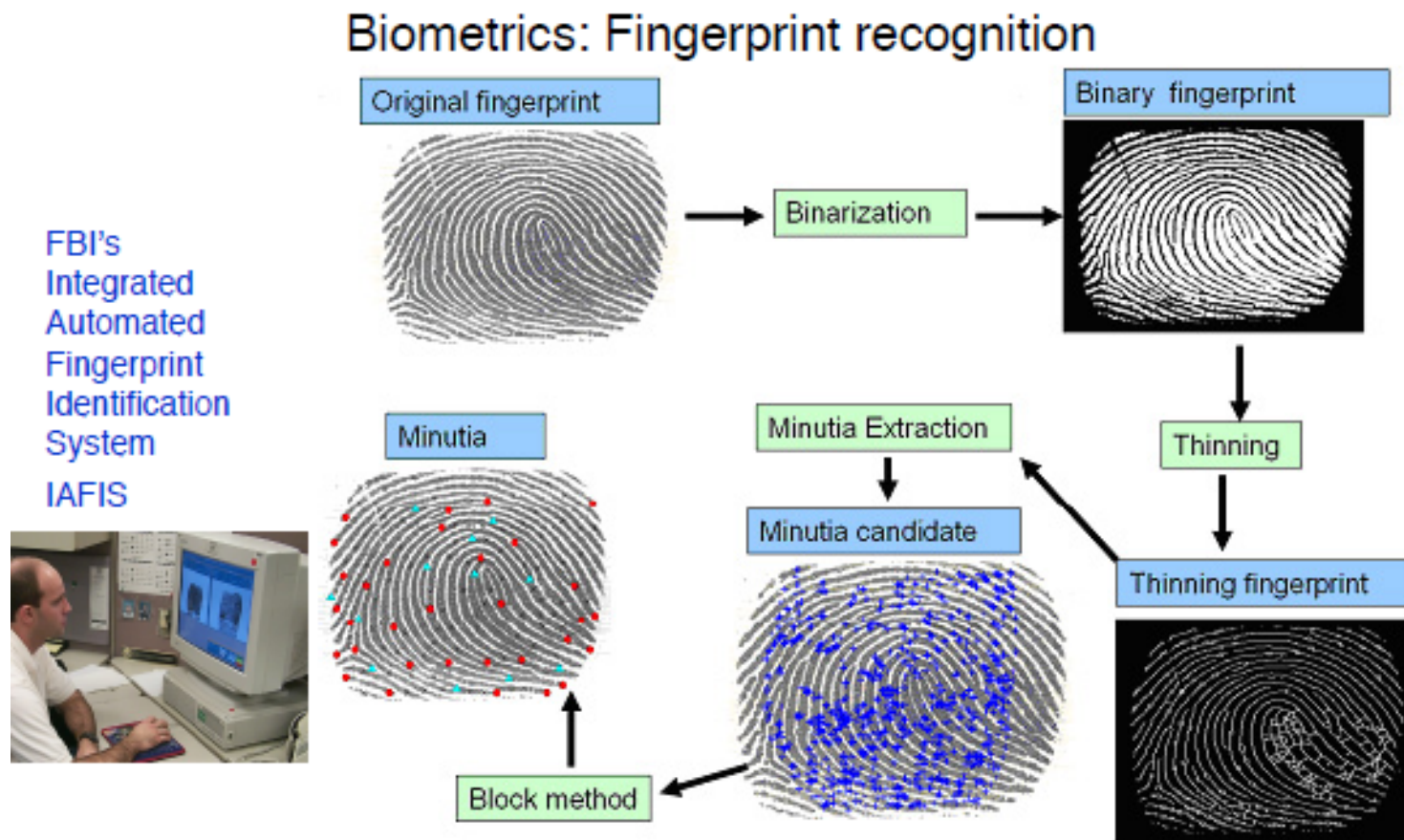
边界及其链码



链码:443212120766665

(c) 8-方向链码

图像匹配与识别



课程特点

- 课程特点

- 对数学理论要求不太高，但数学是强有力的工具
- 对学习而言，章节间关系相对不强；对于解决问题，需要各种方法的结合
- 注重实践，对原理须理解透彻
- 解决问题需要经验和技巧，还需要数学知识

- 理论与实践相结合

- 课堂教学：侧重基本概念和理论知识，要求理解透彻
- 课后作业：侧重解决实际问题，要求融会贯通
 - 针对具体问题，综合运用相关方法
 - 编程实现：Visual C++或者Matlab
 - 实验测试、分析讨论、优化改进

教学要求

- 成绩评定

- 考试：60%（闭卷，允许带1页A4纸资料入场）
- 平时：40%
 - 课堂表现（10分）
 - 每次课前签到。课上点名未到，每次扣2分
 - 书面作业（10分），一般每章布置一次，下周上课前上交
 - 上机作业（20分）
 - 平均两周一次，8个左右；
 - 提交源代码和详细实验报告，阐明目标、原理、方法和结果，包括输入图像数据和结果图像（和数据），编程语言：Visual C++或MATLAB
 - 交电子版报告，布置后的两周内必须上交。
 - 必须亲自做！
 - 鼓励和结合课程内容和自己兴趣，进行一个综合设计，择优展示。视情况总分加1~5分。

- 课件、交流、作业上传

- 学在浙大网站

第一章 概 论

- 图像处理基本概念
- 图像处理的应用
- 图像处理系统的组成

百闻不如一见。

——无名氏

1.1数字图像处理基本概念

- 数字的含义
 - 一幅图像定义为一个二维函数 $f(x,y)$ ， x,y 是空间坐标，幅值 f 为图像在该处的强度或者灰度。当 x,y 和 f 均为有限的离散值时，图像称为数字图像。基本组成元素为像素。
- 数字图像处理的两大主要应用领域：
 - 改善图示信息以便人们解释
 - 为存储、传输和表示而对图像数据进行处理，以便于机器自动处理。
- 是人类视觉的延伸和拓展

图像基本概念

- 人类的感知仅限于电磁波谱的视觉波段，而成像机器几乎可以覆盖从伽马射线到无线电波的整个电磁波谱。

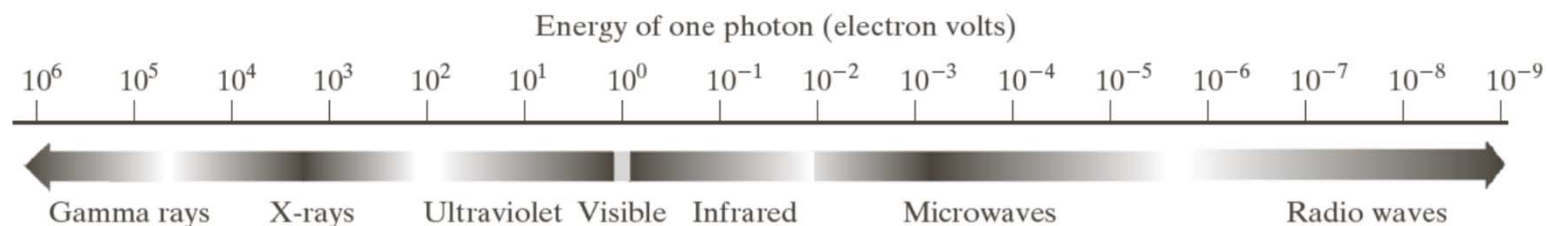


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

人类了解世界是盲人摸象，做人要谦虚！

数字图像的开端

近**100**年（**1921**年）前开始的
第一次大西洋海底电缆图片传输。
可以称为数字图像，但还没有处理。



FIGURE 1.2 A digital picture made in 1922 from a tape punched after the signals had crossed the Atlantic twice. (McFarlane.)

穿孔纸
带还原
图像



FIGURE 1.1 A digital picture produced in 1921 from a coded tape by a telegraph printer with special type faces. (McFarlane.[†])

电报打印机
打印图像



FIGURE 1.3 Unretouched cable picture of Generals Pershing and Foch, transmitted in 1929 from London to New York by 15-tone equipment. (McFarlane.)

15级灰
度图像

数字图像处理的开端

- 晶体管、集成电路的发明，计算机的出现
- 数字图像处理的两个基本需求：
大容量存储和显示系统
- 1964年JPL处理的第一张月球图片
是计算机处理数字图像的开始

FIGURE 1.4 The first picture of the moon by a U.S. spacecraft. *Ranger 7* took this image on July 31, 1964 at 9:09 A.M. EDT, about 17 minutes before impacting the lunar surface. (Courtesy of NASA.)



当前图像处理的应用

- **多媒体通信**：视频会议，可视电话，数字电视，视频监控，视频聊天，IPTV, ...
- **宇宙探索**：太空星体图像的处理...
- **遥感技术**：地球资源调查，气象，污染监测，城市规划，红外测温，GIS
- **生物医学**：放射图像，超声，显微分析，断层扫描、磁共振、PET ...
- **工业生产**：自动化，视觉传感器，无损探伤，金相分析，红外热像测温 ...
- **计算机科学**：文字输入，手势识别，多媒体，广告，搜索，动漫， ...
- **数字娱乐**：DC, DV, DTV, DVR, DVD, PMP, MP4, 电子相框...
- **军事技术**：侦察照片，视频制导，雷达、声纳、红外图像处理、机器人导航...
- **公共安全**：指纹鉴别，印鉴、手迹辨伪，视频监控，视觉监控，步态识别，人眼视线识别 ...
- **智能交通**：电子警察，车牌识别，车型识别和统计，视频线圈，异常事件检测 ...
- **图像安全**：加密，数字版权，防篡改，信息隐藏 ...
- **金融**：清分，假钞识别，

伽马射线成像

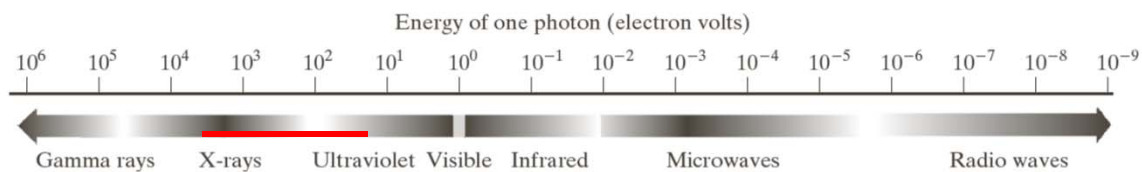
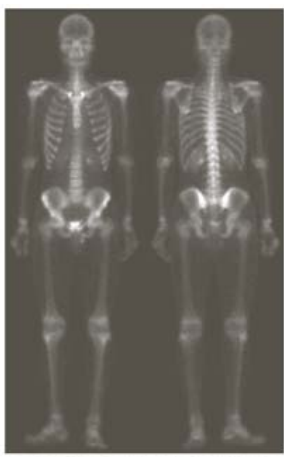


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.



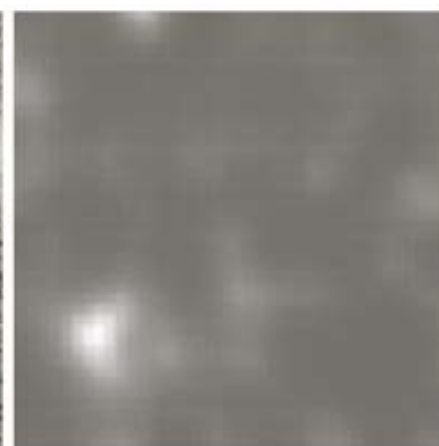
骨骼扫描



PET

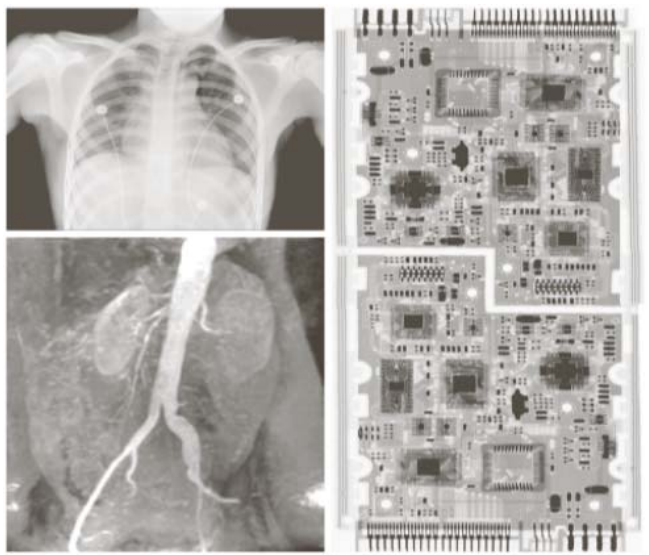


天鵝星座环图像

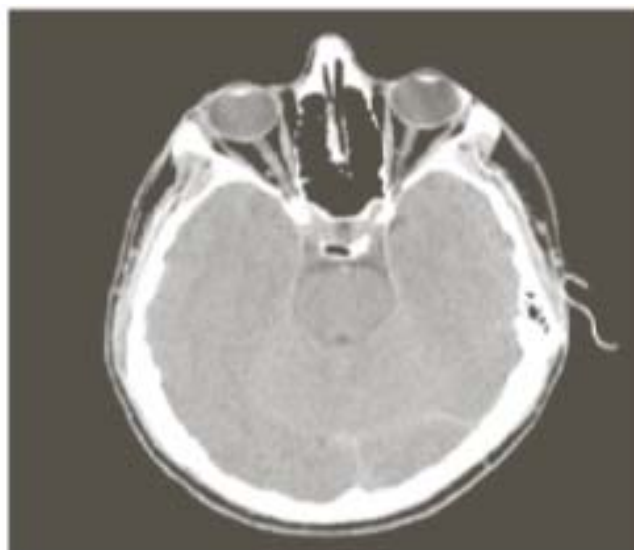


反应堆真空管伽马辐射

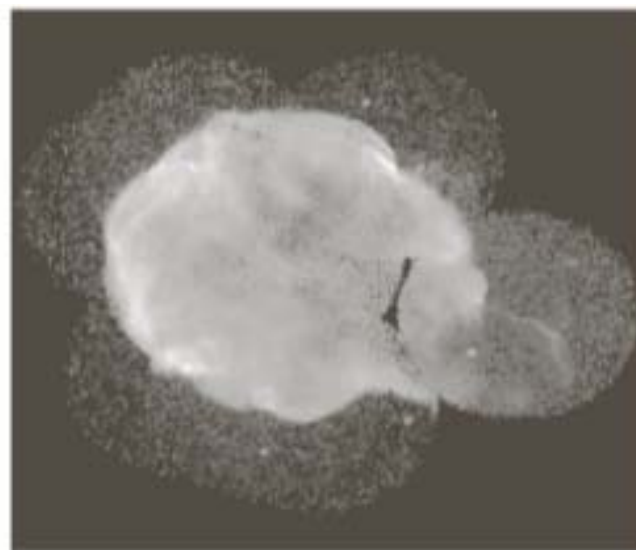
X射线成像



胸部X线，主动脉造影，电路板



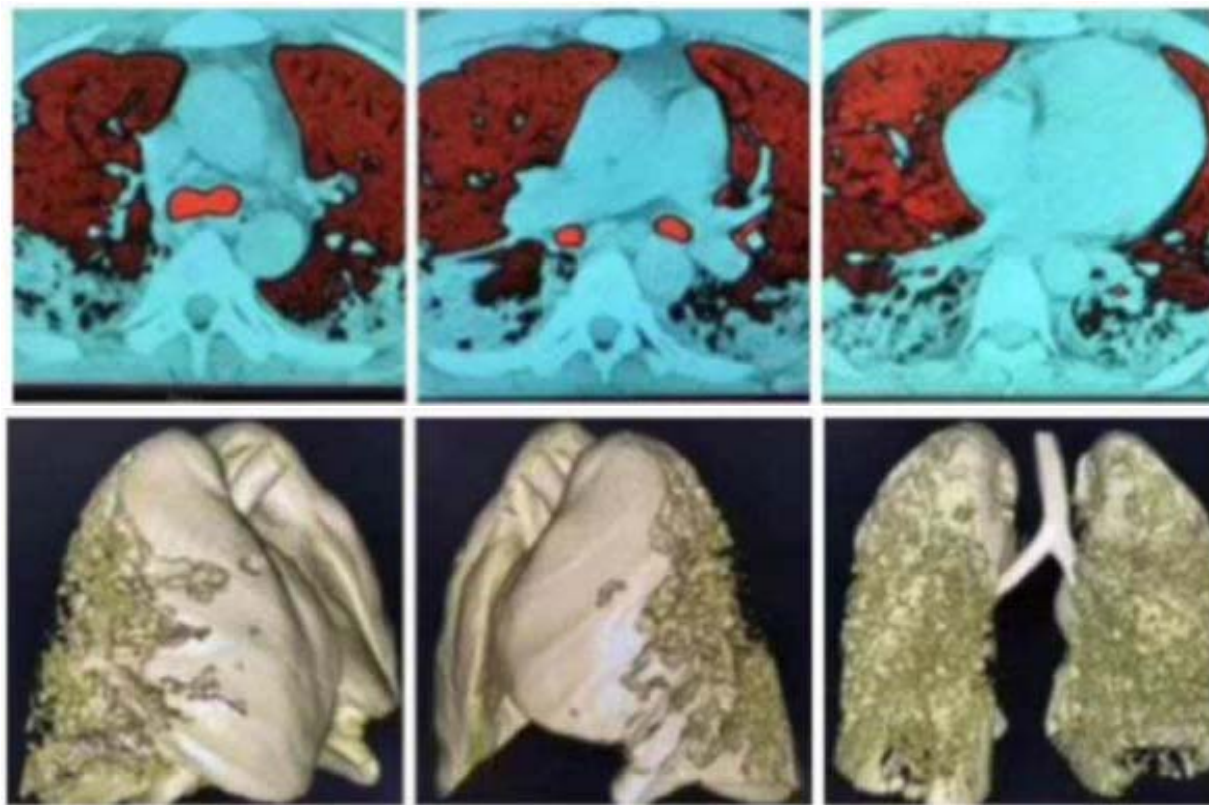
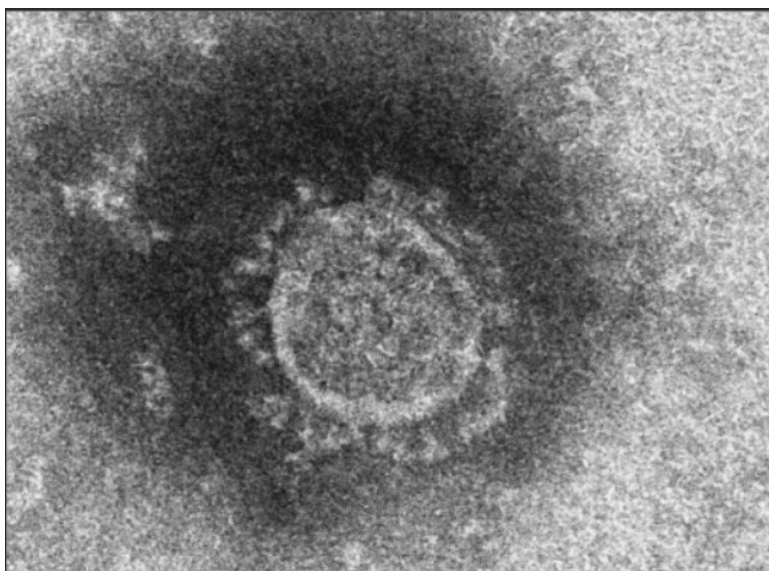
头部CT



天鹅星座环

X射线成像

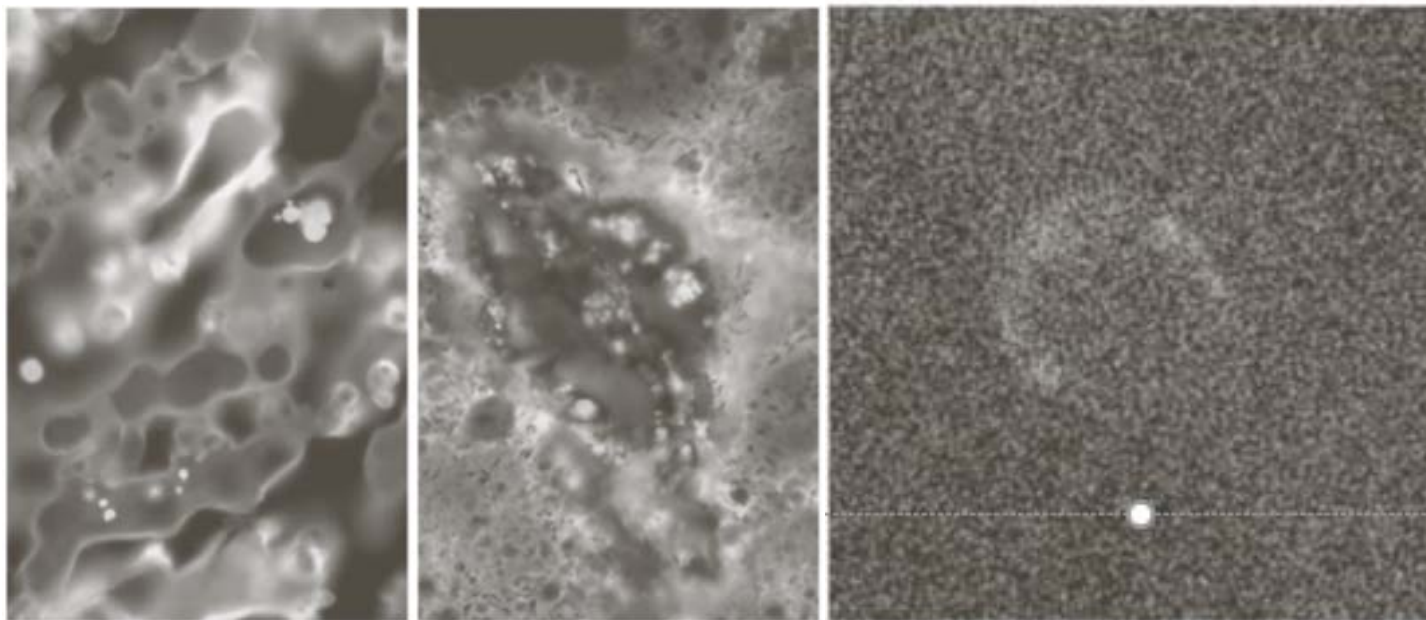
新冠状病毒（下，电子显微镜图像）及其受感染的肺部（伪彩，右，CT）



紫外波段成像

a b
c

FIGURE 1.8
Examples of
ultraviolet
imaging.
(a) Normal corn.
(b) Smut corn.
(c) Cygnus Loop.
(Images courtesy
of (a) and
(b) Dr. Michael
W. Davidson,
Florida State
University,
(c) NASA.)

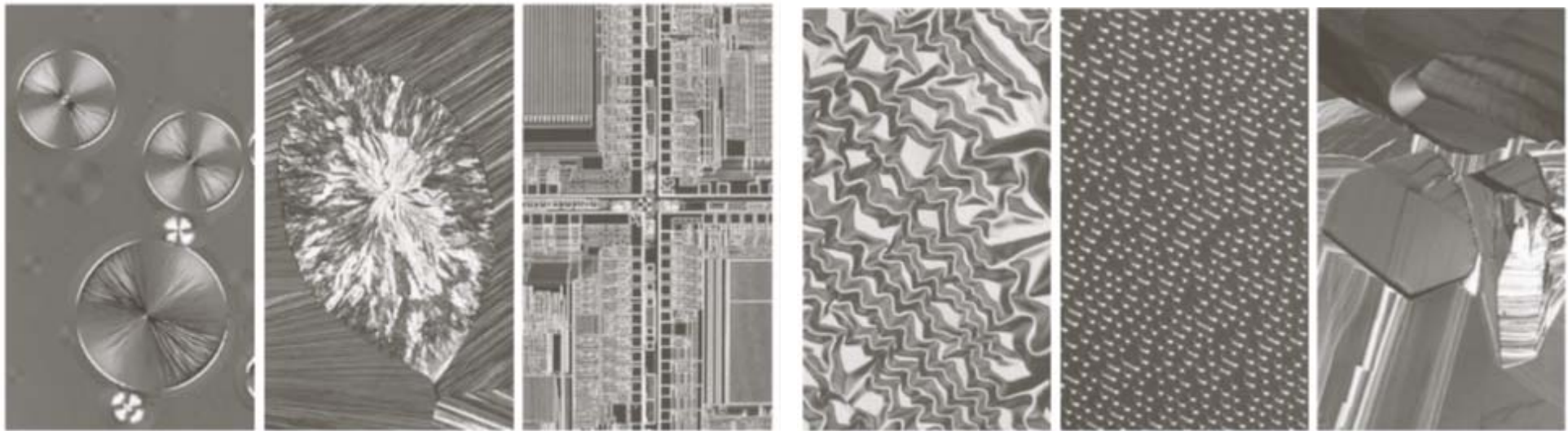


普通玉米

患病玉米

天鹅星座环

可见及红外波段成像I

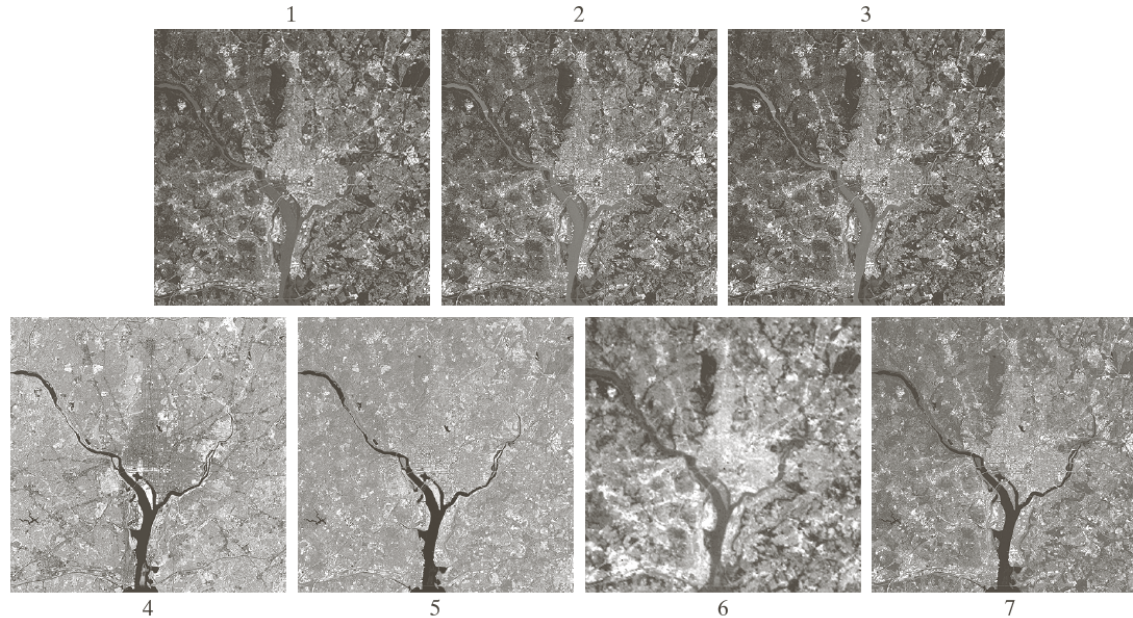


紫杉酚
X250倍

a	b	c
d	e	f

FIGURE 1.9 Examples of light microscopy images. (a) Taxol (anticancer agent), magnified 250 $\times$. (b) Cholesterol—40 $\times$. (c) Microprocessor—60 $\times$. (d) Nickel oxide thin film—600 $\times$. (e) Surface of audio CD—1750 $\times$. (f) Organic superconductor—450 $\times$. (Images courtesy of Dr. Michael W. Davidson, Florida State University.)

可见及红外波段成像



Band No.	Name	Wavelength (μm)	Characteristics and Uses
1	Visible blue	0.45–0.52	Maximum water penetration
2	Visible green	0.52–0.60	Good for measuring plant vigor
3	Visible red	0.63–0.69	Vegetation discrimination
4	Near infrared	0.76–0.90	Biomass and shoreline mapping
5	Middle infrared	1.55–1.75	Moisture content of soil and vegetation
6	Thermal infrared	10.4–12.5	Soil moisture; thermal mapping
7	Middle infrared	2.08–2.35	Mineral mapping

可见及红外波段成像 (10.0~13.4um红外波段)

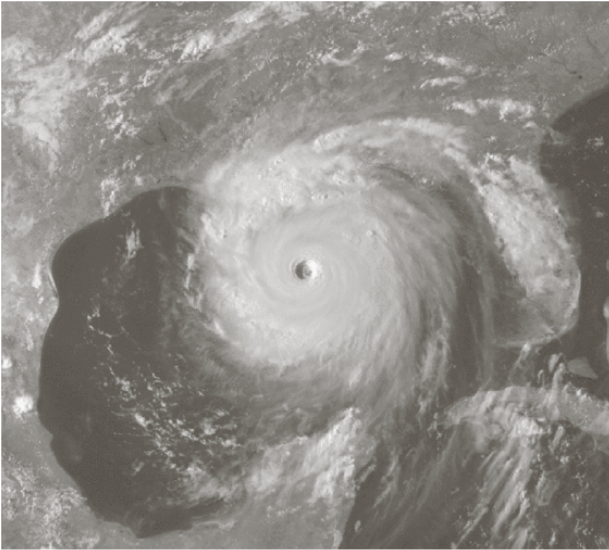


FIGURE 1.11
Satellite image
of Hurricane
Katrina taken on
August 29, 2005.
(Courtesy of
NOAA.)

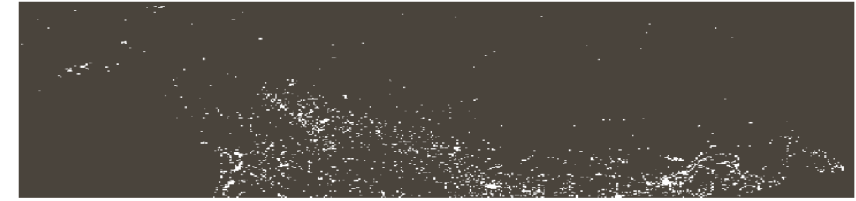
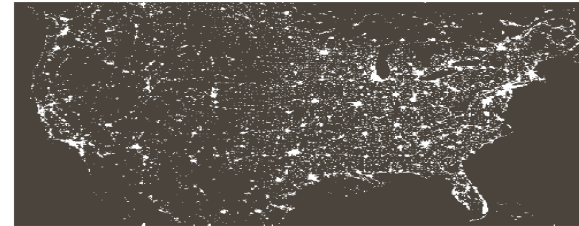


FIGURE 1.12
Infrared satellite
images of the
Americas. The
small gray map is
provided for
reference.
(Courtesy of
NOAA.)



可见及红外波段成像



FIGURE 1.13
Infrared satellite
images of the
remaining
populated part of
the world. The
small gray map is
provided for
reference.
(Courtesy of
NOAA.)

可见及红外波段成像

a b
c d
e f

FIGURE 1.14

Some examples of manufactured goods often checked using digital image processing.

(a) A circuit board controller.

(b) Packaged pills.

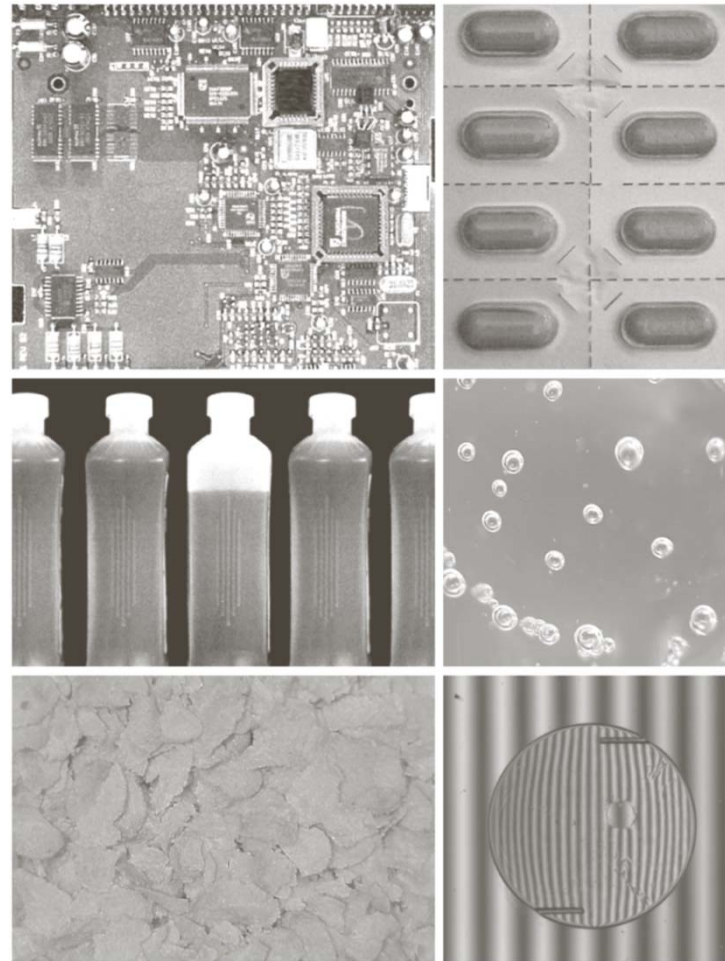
(c) Bottles.

(d) Air bubbles in a clear-plastic product.

(e) Cereal.

(f) Image of intraocular implant.

(Fig. (f) courtesy of Mr. Pete Sites, Perceptics Corporation.)



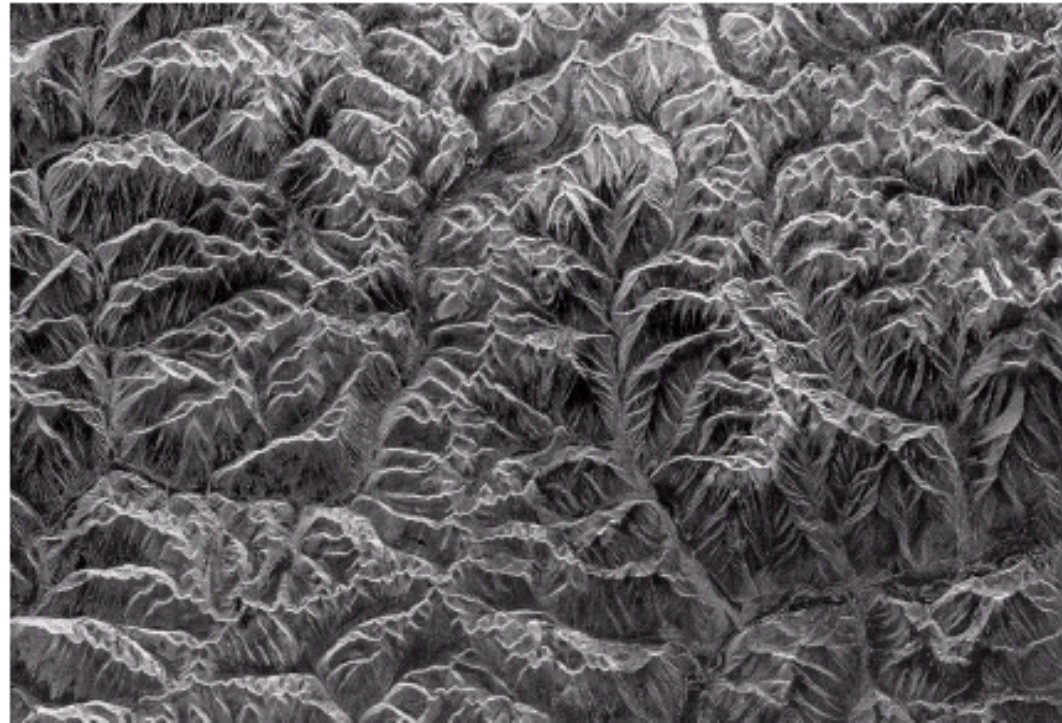
- 可用于产品的自动视觉检测

可见及红外波段成像



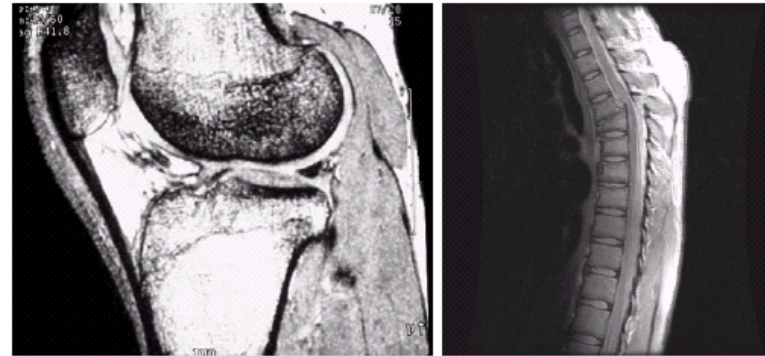
微波波段成像（雷达）

FIGURE 1.16
Spaceborne radar
image of
mountains in
southeast Tibet.
(Courtesy of
NASA.)



- 1. 毫米波雷达的强悍之处在于全天候、全天时的数据采集能力。**
- 2. 某些波段可以穿透云层、植被、冰层和地表。**

无线电波段成像



MRI磁共振图像

a b

FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

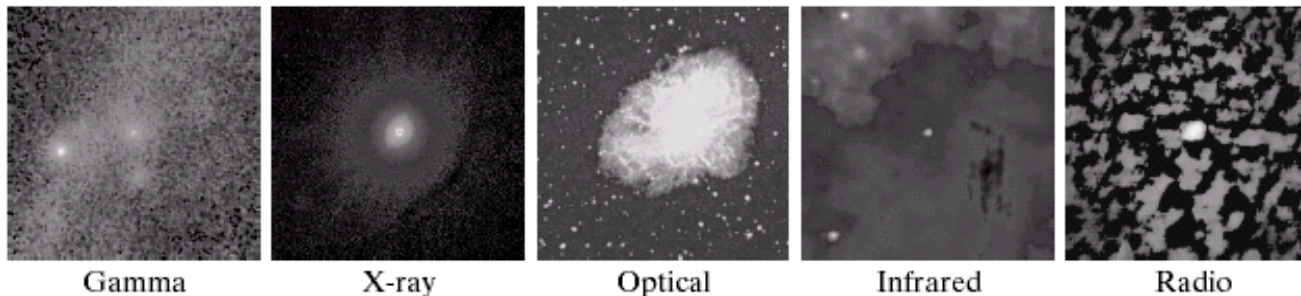


FIGURE 1.18 Images of the Crab Pulsar (in the center of images) covering the electromagnetic spectrum. (Courtesy of NASA.)

覆盖电磁波谱的星云图像

非电磁波谱——声波和超声波成像

低频声波油气勘探 ($\sim 100\text{Hz}$)

FIGURE 1.19
Cross-sectional image of a seismic model. The arrow points to a hydrocarbon (oil and/or gas) trap. (Courtesy of Dr. Curtis Ober, Sandia National Laboratories.)

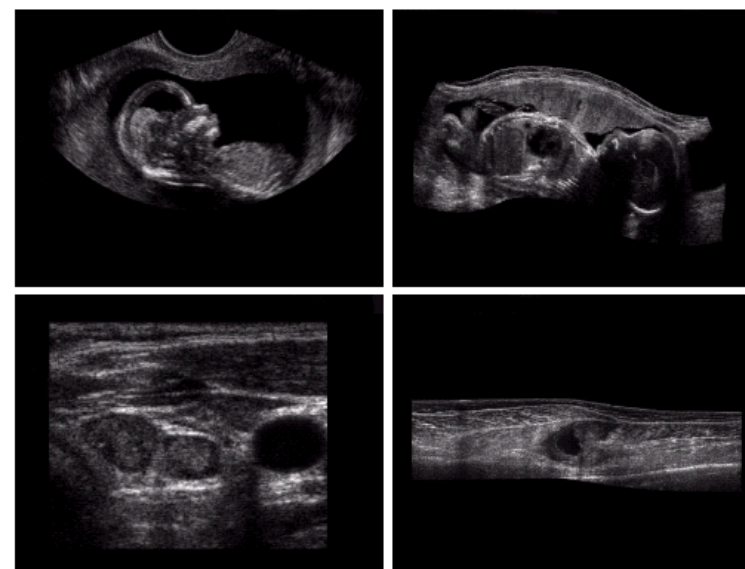
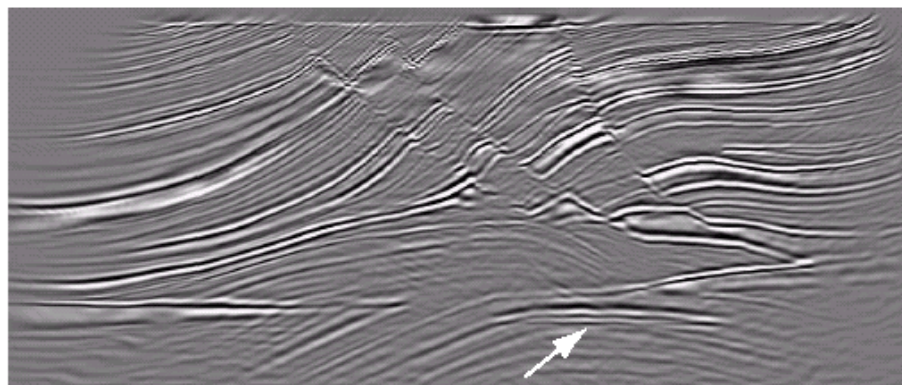
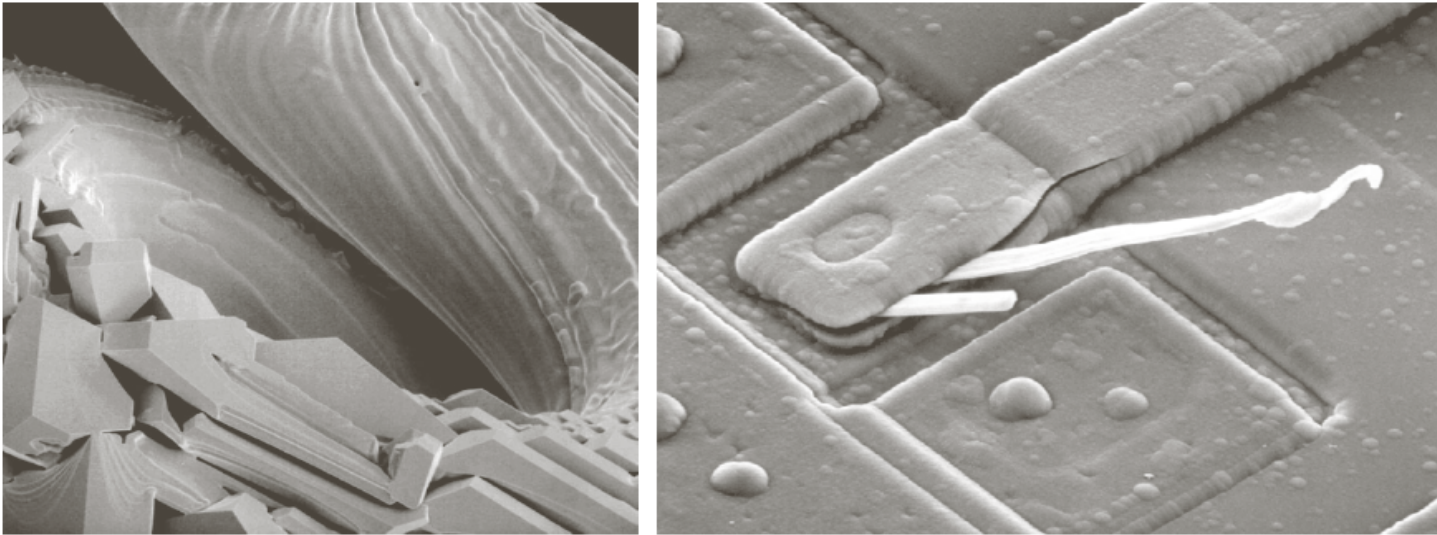


FIGURE 1.20
Examples of ultrasound imaging. (a) Baby. (2) Another view of baby. (c) Thyroids. (d) Muscle layers showing lesion. (Courtesy of Siemens Medical Systems, Inc., Ultrasound Group.)

超声波胎儿检测 ($\sim \text{MHz}$)

透射电子显微镜成像



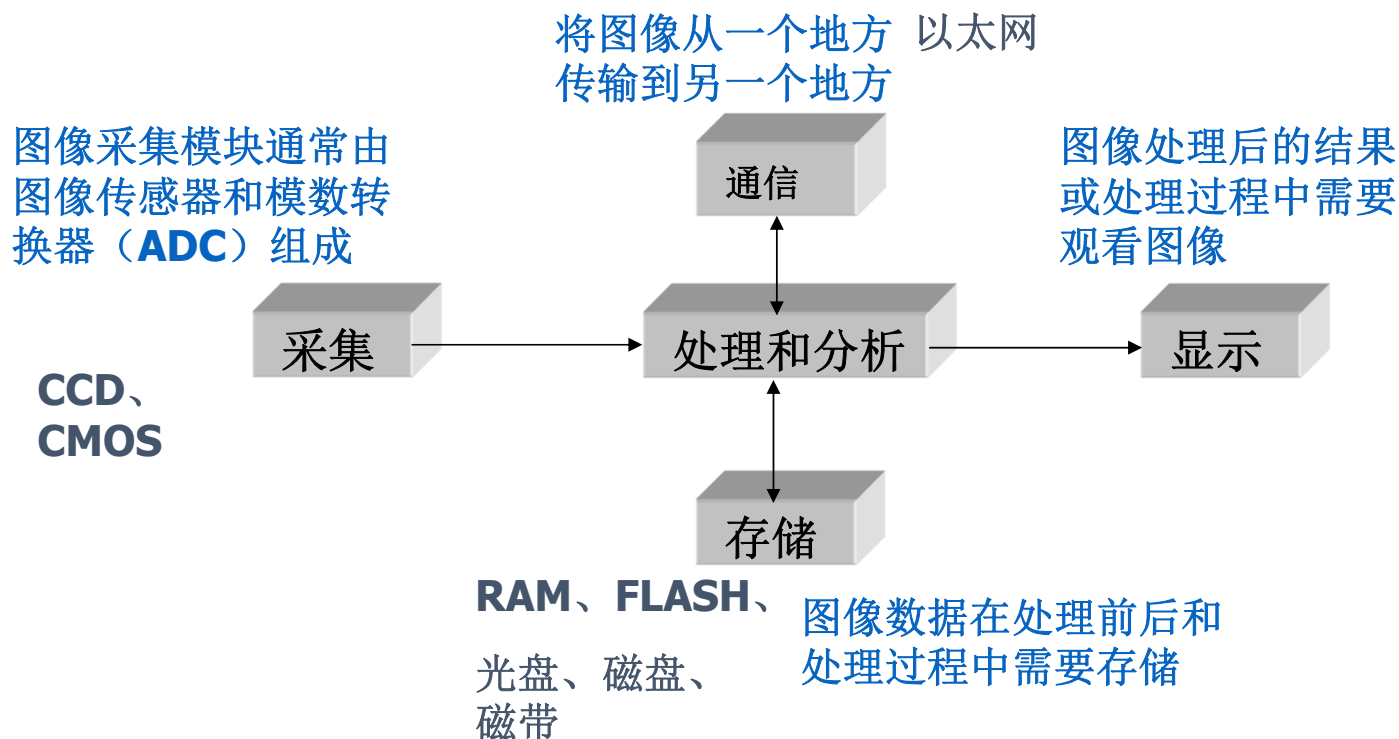
1. 电子显微镜用聚焦的电子波束代替光束形成图像样本
2. 放大倍数可达普通光学显微镜的**10**倍以上

a b

FIGURE 1.21 (a) 250 $\times$ SEM image of a tungsten filament following thermal failure (note the shattered pieces on the lower left). (b) 2500 $\times$ SEM image of damaged integrated circuit. The white fibers are oxides resulting from thermal destruction. (Figure (a) courtesy of Mr. Michael Shaffer, Department of Geological Sciences, University of Oregon, Eugene; (b) courtesy of Dr. J. M. Hudak, McMaster University, Hamilton, Ontario, Canada.)

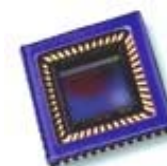
1.3 图像处理系统

- 图像分析处理系统的基本结构



图像采集

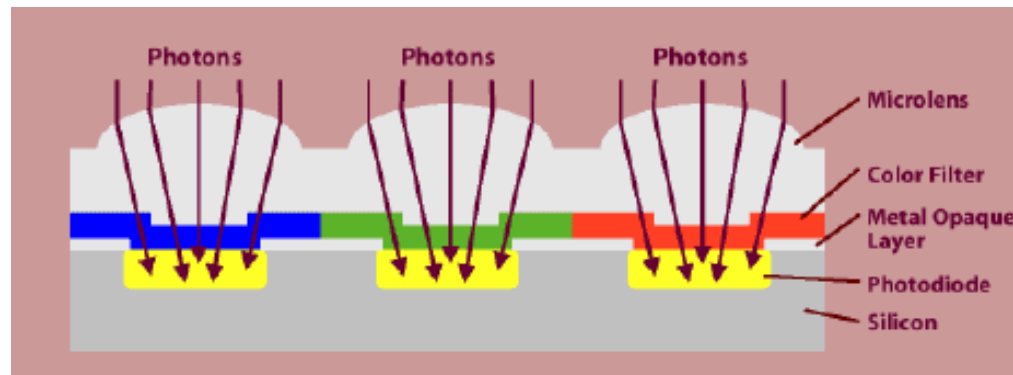
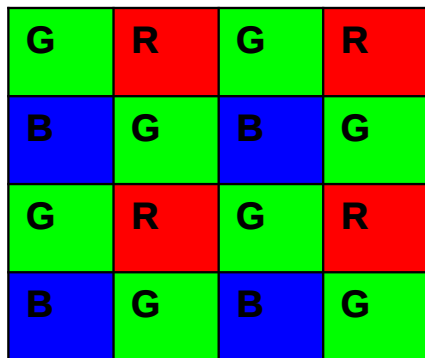
- 图像采集：图像传感器和模数转换器（ADC）
- 目前常用图像传感器件
 - CCD：面阵和线阵两种形式
 - CMOS：集成度更高



其它:红外传感器

彩色CCD传感器

- RGB "Bayer" Color and MicroLenses

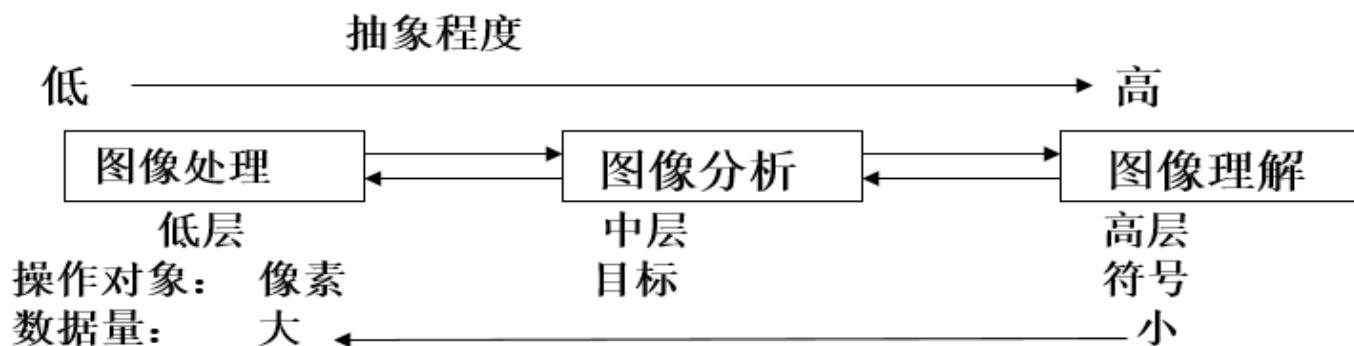


通过内插产生所缺的颜色

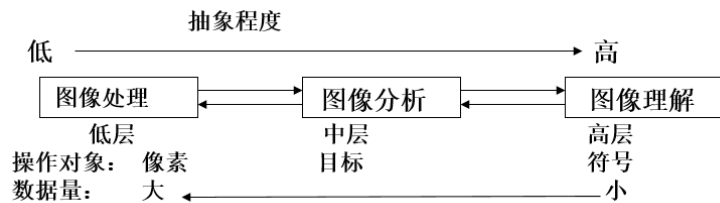
图像处理的实际实现

- 处理和分析
 - 一般先设计好算法，再由软件来实现
 - 对速度要求高的场合，也采用专门适合于图像处理的硬件来实现（如流水线结构、阵列结构、并行结构、专用部件等）。
 - 处理器：
 - 通用的处理器（如PC机，工作站）
 - 数字信号处理器（DSP）
 - 具有并行视频处理功能的DSP、GPU
 - ASIC：专为实现某种图像处理算法（如图像压缩）设计

图像工程的三个层次



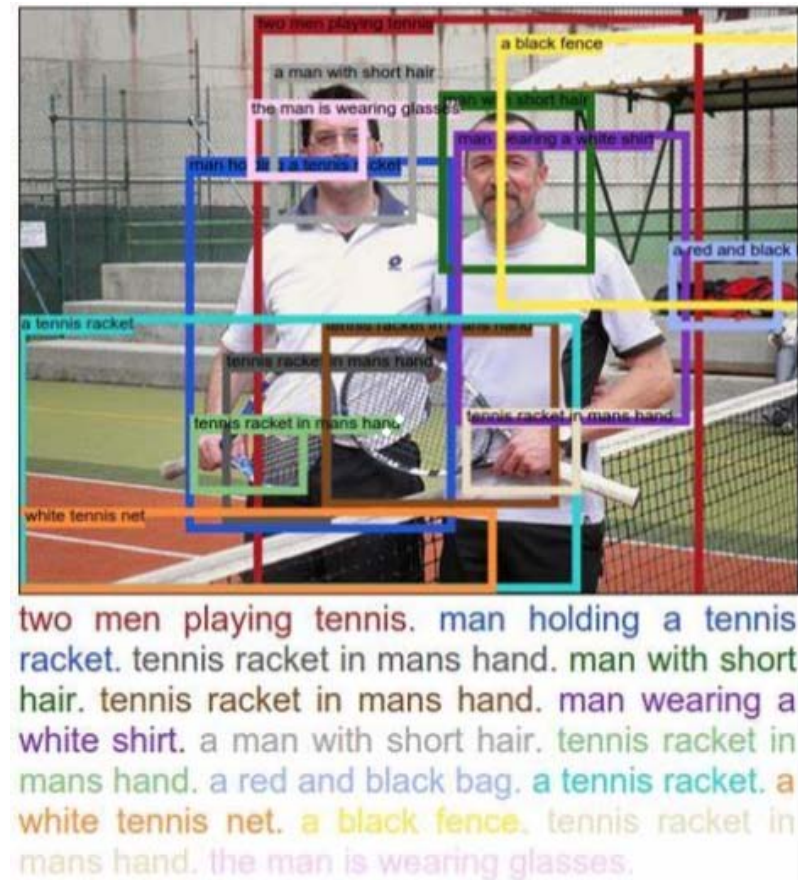
- 基本图像处理（图像预处理）
 - 如，图像增强以改善视觉效果，模糊图像的恢复，图像的解压缩处理，图像去失真
- 图像分析
 - 主要是对图像中感兴趣的目标进行检测（分割）和测量（特征提取），提取出对它进行描述的信息。
 - 结果：可以是对图像描述的特征数据（如长度、面积等），或用符号表示的特征



• 图像理解 (计算机视觉)

- 在图像分析的基础上, 找出各目标的性质和它们之间的相互联系, 得出对图像内容和含义的理解以及对客观场景的解释 (语义理解)

- 典型应用: 无人驾驶车, 机器人...



图像理解（几何理解）

- 场景立体重构（三维）
 - 多视角，立体视觉
 - 深度信息——深度图（Depth Map）
- 从单幅图像恢复深度（深度网络）

